

Chapter 1

Introduction

Every day, we interact with the environment naturally through our skin. It happens in sports, during a shopping day or in other activities. Most of the time, we are affected by stress because obliged tasks start to accumulate day-by-day.

Responding to a clear need to cope with this issue, research recently advances into the area of on-body interaction. Using modern electronics and data around us, devices for the skin have been developed to assist the users with their daily life activities 1.1. The body can be used as an input surface to enable distinct interaction possibilities e.g. accepting calls by touching the ear. In addition, the commonly known touch input can be extended with complex hand gestures that the human body naturally affords which opens up more intuitive approaches e.g. tap, pinch or rub. Many of these projects aim to create an always-available, intuitive and supportive technology that improves the way we engage in our environment. For instance, a visionary challenge is to equip the human body with a small processing sensor to facilitate direct access to a users' biomedical signals [1]. In an unforeseen car accident, the ambulance could immediately retrieve the medical documents from an unconscious victim by using his sensors. Furthermore, they can also be integrated as colorful overlays for the human skin which appeals to the aesthetics of body decorations e.g. sensors that look like tattoos [2, 3].

However, the fabrication method of such sensors vary highly and can demand a lot of costs since the design overlay is often applied manually. The human body has unique properties, hence an analog recalibration procedure has to be included for every new user. When mistakes happen during the fabrication, devices often have to be built from scratch again which can be expensive and time-consuming i.e. positioning, scaling, copying or reorienting a 2D pattern repetitively on the body is a stagnant analog process.



Figure 1.1: An example of different on-skin sensors. Figure acquired ...

As new forms of 3D printing and digital fabrication emerges, it is important to provide automatized tools that are capable to ease this process for us. In more detail, it is necessary to have a mean to create designs that are easily replicable, shareable with others, archivable, fabricated remotely, and adaptable to different bodies. [ref] Additionally, there is a potential market to attract the audience - without the technical background - to design and customize their own products. In various cases, such personalization is especially interesting for producing objects which adapt perfectly on a user's body e.g. braces, jewelry and other wearable devices.

Since the aforementioned idea is a recent discovery in human-computer interaction (HCI) very few design tools have been developed. It is hence important to introduce the first step towards the larger vision of easy customization and prototyping.

1.1 Our approach

We propose a digital design application (subsequently referred to as Scansify) which enables users to create a layout and then fabricate on-skin sensors [4] that conforms to the user's body. Using a handheld depth sensor camera, our digital framework scans the outer form of the human body and allows the user to digitally draw on the reconstructed model. After the drawing process, the input forwards into a post-processing pipeline and converts to a flat 2D shape which can be used for creating sensors [ref to pipeline chapter]. After fabricating, the overlay wraps perfectly on the user's arm as designed in the tool 1.2.



Figure 1.2: An example of the created 2D shape. Figure acquired ...

- a) Drawing the layout in 3D;
- b) Our tool outputs a 2D shape which can be further processed;
- c) The shape fits perfectly on the skin



Scansify provides guidance, feedback, and support of **additional operations**. The following list summarizes the key implemented contribution of this thesis:



Core: **The** main functionality

- a) Capturing the human body with a RGB-D camera for acquiring precise geometry.
- b) Live display of skin reconstruction process and depth stream from the camera
- c) Ability to annotate the digital body with a graphical user interface.
- d) Unwrap algorithm that transforms the 3D drawing into a 2D shape.



Sub: The remaining functionality that improves usability



- e) Import and **Export** of designs and 3d models.
- f) Revert of latest change or completely reset the design
- g) Camera **view can be manipulated** such as steering, zooming or rotating.



Our approach thrives to target users that **might or might not have the skills to create on-body sensors** by providing an intuitive product for rapid prototyping. At the end, a use-case study is conducted and results are discussed to evaluate the

feasibility and usability of the product. In addition, feedback is taken to refine the design principles for future iterations.

1.2 Structure of the thesis

The following list depicts a short overview of the subjects covered in the remaining chapters:

Chapter 2



presents recent work in on-body interaction and depth-sensing **advances**.

Chapter 3



introduces **concept and novel design considerations** as a result of the related work chapter.

Chapter 4

illustrates a detailed workflow description of the graphical user interface and outlines the development process of Scansify.

Chapter 5



evaluates the results from a user study **with respect to** the feasibility and usability of our tool.

Chapter 6



concludes the thesis with a discussion **over limitation** and approaches for future work.



References

- [1] T. S. Saponas, D. S. Tan, D. Morris, R. Balakrishnan, J. Turner, and J. A. Landay. “Enabling Always-available Input with Muscle-computer Interfaces”. *Proceedings of the 22Nd Annual ACM Symposium on User Interface Software and Technology*. UIST '09. Victoria, BC, Canada: ACM, 2009, pp. 167–176. DOI: 10.1145/1622176.1622208. URL: <http://doi.acm.org/10.1145/1622176.1622208>.
- [2] J. Lo, D. J. L. Lee, N. Wong, D. Bui, and E. Paulos. “Skintillates: Designing and Creating Epidermal Interactions”. *Proceedings of the 2016 ACM Conference on Designing Interactive Systems*. DIS '16. Brisbane, QLD, Australia: ACM, 2016, pp. 853–864. DOI: 10.1145/2901790.2901885. URL: <http://doi.acm.org/10.1145/2901790.2901885>.
- [3] H.-L. C. Kao, C. Holz, A. Roseway, A. Calvo, and C. Schmandt. “DuoSkin: Rapidly Prototyping On-skin User Interfaces Using Skin-friendly Materials”. *Proceedings of the 2016 ACM International Symposium on Wearable Computers*. ISWC '16. Heidelberg, Germany: ACM, 2016, pp. 16–23. DOI: 10.1145/2971763.2971777. URL: <http://doi.acm.org/10.1145/2971763.2971777>.
- [4] A. S. Nittala, A. Withana, N. Pourjafarian, and J. Steimle. “Multi-Touch Skin: A Thin and Flexible Multi-Touch Sensor for On-Skin Input”. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. CHI '18. Montreal QC, Canada: ACM, 2018, 33:1–33:12. DOI: 10.1145/3173574.3173607. URL: <http://doi.acm.org/10.1145/3173574.3173607>.